

VINNYTSIA, Ukraine

WMO: 335620

Lat: 49.2478N

Lon: 28.6042E

Elev: 978

StdP: 14.18

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	4.7	-6.7	4.0	-0.4	-0.5	5.5	5.6	26.9	31.4	23.1	34.3	6.9	0	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	86.3	67.5	83.2	66.4	80.5	65.2	70.1	81.6	68.6	79.6	67.1	77.4	8.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	100.4	75.3	64.7	94.8	73.4	63.3	90.1	72.0	34.6	81.9	33.3	79.8	32.1	77.6	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.4	18.4	16.4	DB	-7.5	90.2	7.0	3.4	-12.5	92.7	-16.6	94.7	-20.5	96.6	-25.6	99.1	(4)
(5)			WB	-7.9	72.4	6.9	1.9	-12.9	73.7	-16.9	74.8	-20.8	75.9	-25.8	77.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	24.3	27.0	35.5	48.6	58.7	64.5	67.9	66.7	57.4	46.7	37.1	27.8	(6)	
	DBStd	17.48	10.15	10.21	8.64	7.68	7.26	6.08	5.45	6.03	6.97	7.93	8.64	9.68	(7)	
	HDD50	3275	796	645	452	114	14	0	0	0	15	157	393	689	(8)	
	HDD65	6922	1261	1065	915	493	219	84	33	51	243	567	837	1154	(9)	
	CDD50	2162	0	0	2	72	282	435	556	517	237	55	6	0	(10)	
	CDD65	333	0	0	0	1	22	69	124	103	14	0	0	0	(11)	
	CDH74	2876	0	0	0	16	208	534	1020	952	143	3	0	0	(12)	
	CDH80	702	0	0	0	2	29	91	273	283	24	0	0	0	(13)	
(14) Wind	WSAvg	8.7	9.3	9.7	9.8	9.3	8.4	8.0	7.6	7.2	7.9	8.4	9.3	9.3	(14)	
(15) Precipitation	PrecAvg	24.9	1.5	1.4	1.3	1.8	2.3	3.2	3.9	2.3	2.0	1.5	1.8	1.6	(15)	
	PrecMax	33.7	5.9	3.6	3.3	4.7	5.3	7.6	9.7	4.9	7.2	3.9	5.9	3.1	(16)	
	PrecMin	16.9	0.3	0.3	0.1	0.4	0.5	0.6	0.5	0.6	0.0	0.1	0.2	0.3	(17)	
	PrecStd	3.9	1.0	0.9	0.8	1.1	1.2	1.9	2.0	1.3	1.7	1.1	1.1	0.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	50.9	61.3	75.5	83.6	85.8	90.7	90.8	82.8	72.6	60.7	49.9	(19)	
		MCWB	41.1	44.4	49.4	57.9	63.5	67.4	68.9	68.8	63.1	59.2	52.7	46.2	(20)	
	2%	DB	41.0	44.7	56.0	69.9	79.4	82.7	86.5	86.5	78.2	67.4	55.1	45.4	(21)	
		MCWB	38.6	40.9	46.2	54.6	61.8	66.4	68.0	67.5	61.3	57.2	50.5	43.1	(22)	
	5%	DB	38.1	41.5	51.5	65.5	75.7	80.1	82.8	82.8	74.2	62.2	51.7	42.3	(23)	
		MCWB	36.1	38.7	43.7	52.3	60.9	65.4	67.4	66.1	60.2	54.8	49.1	40.6	(24)	
	10%	DB	35.8	38.3	47.4	62.0	72.1	77.0	79.7	79.4	70.0	58.2	48.7	39.1	(25)	
		MCWB	34.3	36.1	41.7	50.4	58.9	64.3	66.0	65.0	58.8	52.6	46.4	37.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	45.0	50.5	58.7	66.4	71.5	71.9	71.9	66.4	61.5	54.3	47.3	(27)	
		MCDB	43.8	49.0	59.4	73.3	78.2	80.5	83.6	84.6	78.1	69.6	58.5	49.3	(28)	
	2%	WB	38.9	41.2	47.3	56.3	64.1	68.9	70.2	69.7	63.3	58.3	51.4	43.3	(29)	
		MCDB	40.7	44.0	54.0	67.6	75.6	78.8	81.8	81.9	73.6	64.7	54.2	45.1	(30)	
	5%	WB	36.4	38.9	44.5	53.7	62.1	67.2	68.8	67.9	61.3	55.6	49.1	40.7	(31)	
		MCDB	37.8	41.3	49.9	62.8	72.8	76.6	79.9	79.5	70.7	61.1	51.4	42.1	(32)	
	10%	WB	34.4	36.3	42.1	51.6	60.1	65.4	67.4	66.2	59.5	53.1	46.6	37.6	(33)	
		MCDB	35.5	38.0	46.8	59.5	69.8	74.3	77.5	76.7	68.2	57.4	48.7	38.9	(34)	
(35) Mean Daily Temperature Range	MDBR		7.9	9.1	12.3	16.5	18.1	17.2	17.7	18.6	16.3	13.7	8.8	7.8	(35)	
	5% DB	MCDBR	8.0	11.8	19.4	22.1	22.4	21.2	22.5	23.5	23.2	20.1	13.5	9.8	(36)	
		MCWBR	7.0	8.9	11.6	10.7	9.6	9.0	8.4	9.1	10.2	11.3	9.1	8.1	(37)	
	5% WB	MCDBR	7.6	11.2	17.5	20.5	20.4	19.0	19.9	20.9	19.4	17.9	11.6	9.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.308	0.338	0.375	0.434	0.404	0.401	0.429	0.430	0.392	0.360	0.339	0.307	(40)	
	taud		2.234	2.186	2.203	2.112	2.272	2.298	2.230	2.221	2.291	2.357	2.327	2.292	(41)	
	Ebn at Noon		230	250	261	257	272	273	262	255	253	242	217	214	(42)	
	Edh at Noon		24	32	37	46	41	40	42	41	34	27	22	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		305	553	958	1400	1817	1922	1848	1635	1123	682	291	212	(44)	
	RadStd		57	106	136	147	146	133	117	95	167	101	30	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.40	N/A	N/A	+1.55	+1.19	+1.01	-297	-436	+245	+90
Regional (11 neighbors)		+1.30	N/A	N/A	+1.62	+0.88	N/A	-245	-364	+214	+92

Nomenclature: See separate page